Student Performance Analytics & Prediction

A beginner-friendly Data Analytics and Machine Learning project

Prepared for: Internship project portfolio
Technology stack: Python, Pandas, Matplotlib, Seaborn, scikit-learn, Streamlit

Dataset: 240 synthetic sample records created for demonstration. No real student personal data is used.

1. Abstract

This project explores student performance data and demonstrates a complete introductory analytics workflow. It summarizes data, visualizes score patterns, and trains a Random Forest regression model to estimate a final score from study habits and academic indicators. A Streamlit dashboard makes the analysis accessible through a simple web interface.

2. Problem Statement

Students and educators may want to understand how study time, attendance, assignment completion, sleep, previous scores, and internet access relate to academic outcomes. This project demonstrates how data analytics and machine learning can help explore these relationships. It does not claim that these factors cause a particular score.

3. Objectives

1) Load and inspect a structured CSV dataset. 2) Calculate descriptive statistics. 3) Visualize score distribution and numeric correlations. 4) Train and evaluate a regression baseline. 5) Present the analysis and an illustrative prediction in a dashboard.

4. Dataset Description

The dataset contains 240 synthetic records. Columns include student_id, study_hours_per_day, attendance_percent, assignments_completed_out_of_10, sleep_hours, internet_access, previous_exam_score, and final_score. The sample data is generated for this project and is not evidence about real students.

5. Methodology

The CSV is loaded with Pandas and checked using summary statistics. Matplotlib and Seaborn are used for a score distribution plot and a correlation heatmap. The dataset is split into training and test sets (80% / 20%). A Random Forest Regressor is trained using study hours, attendance, completed assignments, sleep hours, previous score, and a binary encoding of internet access. Mean Absolute Error (MAE) and R-squared are reported on the held-out test split.

6. Evaluation

The application calculates evaluation metrics when run. MAE expresses the average absolute prediction error in score points; lower is better. R-squared describes how much score variation is explained on the test split, but can be negative when predictions are worse than a simple mean baseline. Results may vary if the dataset or model settings change.

7. Dashboard Features

The Streamlit dashboard provides a dataset preview, descriptive statistics, a score distribution chart, a correlation heatmap, and a form for entering example characteristics to receive an illustrative predicted score.

8. Limitations and Responsible Use

The dataset is synthetic and designed for a learning demo. Its patterns are artificial and may not generalize to real learners. Correlation does not establish causation. A prediction should not be used to grade, rank, admit, discipline, or otherwise make decisions about real students. Real-world deployment would require representative, consented data, privacy safeguards, fairness testing, and domain review.

9. Future Scope

Potential improvements include adding data validation, comparing linear regression and tree-based models, using cross-validation, tracking experiments, adding unit tests, and evaluating the system on an appropriate real dataset with permission and privacy safeguards.

10. How to Run

Install Python 3.10 or later, create a virtual environment, install packages from requirements.txt, and run `streamlit run app.py`. For terminal-based analysis, run `python analysis.py`. See README.md for exact commands.